Rudra Patel

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Education

LDRP Institute of Technology and Research Oct 2021 – May 2025
Bachelor of Engineering in Computer Engineering (CGPA: 7.58) *Gandhinagar, Gujarat*

Technical Skills

Languages & DB: SQL, HTML, CSS, JavaScript, Node.js, Python, Go, Rust, PostgreSQL, MySQL, MongoDB
Frameworks: Tauri, Django, Flask, Streamlit, Express.js
Libraries: React.JS, ChartJS, Plotly, Pandas, NumPy
Tools: Git/GitHub, Docker, Postman, Trello, Jupyter Notebook

Experience

Electrosine Technologies Nov 2025 – Present
Full-Stack Developer *Gandhinagar, Gujarat*

- Developed a scalable Go (Fiber) backend and REST APIs for diagnostic data using SQLC and PostgreSQL. Engineered a zero-downtime database migration to TimescaleDB with a custom 16-bit binary decoder, dynamically routing legacy requests via AWS S3 to reduce latency by 80% with 100% backward compatibility.
- Engineered the frontend for a multi-tenant web portal (React, TypeScript, Tailwind) to monitor distributed motor fleets across facilities. Created modular dashboards (Plotly, Recharts) for 6-hour electrical trends, historical sparklines, and 14-day baseline thresholds, plus admin modules for customers, facilities, equipment nameplates, and power-transmission sequences with debounce logic and role-based access control.
- Architected a cross-platform desktop app (Tauri, React, TypeScript) for offline ESA/MCSA analysis. Built interactive Plotly visualizations for real-time 3-phase waveforms, demodulated currents, and FFT spectrums with fault-frequency overlays. Managed sidecar processes for embedded binary communication, hardware config flows, and automated client-side DOCX reporting.
- Maintained an embedded Rust backend for high-frequency telemetry and on-device ESA, calculating fault frequencies for bearings, gearboxes, eccentricity, and rotor bars. Handled local persistence and updates via SQLite migrations and device-side data pools.

Prodigy InfoTech | **Link** Sept 2024 – Oct 2024
Data Science Intern *Remote*

- Implemented a decision tree classifier (85% accuracy) for marketing data and modeled social media sentiment shifts (82% accuracy). Visualized categorical features in the Titanic dataset to identify a 74% demographic survival rate, and mapped traffic accident hotspots to reveal primary triggers (70% poor roads, 60% night).

Projects

TaskMaster — Team Project Management Platform | *Go*, *React*, *TypeScript*, *PostgreSQL*, *Docker* | GitHub Oct 2026

- Established a layered Go/chi REST API and React client, with PostgreSQL migrations for accounts, workspaces, projects, tasks, comments, labels, and activity logs. Secured account and session flows with bcrypt, short-lived JWT access tokens, rotating refresh tokens in HttpOnly cookies, per-client rate limits, and strict request validation; added protected routes, session restoration, and silent token renewal.

Expense Tracker Web Application | *Python*, *Django*, *PostgreSQL*, *HTML*, *CSS*, *JS* | GitHub Mar 2025

- Designed and enhanced a full-stack expense management system with user authentication and CRUD functionality. Utilized PostgreSQL as the backend database for efficient data storage and retrieval. Generated monthly and yearly expense summaries using Django ORM and query optimization. Built a responsive front-end with Bootstrap to ensure compatibility across devices.

Hand Gesture Recognition System | *Python*, *OpenCV*, *MediaPipe*, *Streamlit* | GitHub Dec 2024

- Implemented a real-time hand gesture recognition system using Python, OpenCV, and MediaPipe, achieving 97% accuracy. Trained custom ML classifiers for static and dynamic gestures, processing 19 3D landmarks at 30+ FPS. Created an interactive web interface using Streamlit for real-time parameter tuning and gesture training. Enabled gesture-based controls, optimizing performance for smooth real-time processing.

Facial Recognition with Attendance System | *Python*, *Tkinter*, *OpenCV*, *MySQL* | GitHub Jan 2024

- Engineered a real-time face recognition system with 90%+ accuracy, leveraging OpenCV and the face recognition library. Used a Tkinter-based GUI for effortless user registration and image storage. Integrated MySQL database for attendance tracking, automatically logging timestamps. Optimized image processing to recognize faces in under 2 seconds.

Certificates

Python And Django Framework For Beginners Complete Course | *Udemy* May 2025

- A comprehensive course on Python and Django, covering core programming, OOP, and full-stack web development.

HackerRank - SQL | *HackerRank* Apr 2025

- Achieved Verified Certification in SQL Basic, Intermediate and Advanced by HackerRank.

NPTEL Database Management System | *NPTEL* Mar 2024

- Secured Elite Category certification in Database Management Systems through NPTEL Exam.